TryHackMe Walkthroughs: Snort Lab Challenge

Today's 'essay' is less of a report and more of a walkthrough of the lab.

NOTE: All the answers to this lab were found by me. The screenshots were also done by me - I used the Mac Screenshot tool (Cmd-Ctrl-Shift-4).

I used Google to figure out how to write the rules and what they meant. *That's all I used it for.*

IDS Rules for HTTP traffic

Answer the questions below

1. Write a rule to detect all TCP packets **from or to** port 80.
   What is the number of detected packets you got?
   Note: You must answer this question correctly before answering the rest of the questions.
   Step 1: Open the 'Local Rules' file
   Step 2: Inside 'Local Rules', `alert tcp any any <> any 80 (msg: "HTTP Port 80 Activity Detected"; sid: 100001; rev:1;)` (this rule tells an IDS to keep an eye on all basic web traffic and alert whenever it sees anything moving across port 80.)
   Step 3:Inside Terminal, `snort -c local.rules -A full -l . -r mx-3.pcap`
   164

```
ubuntu@ip-10-67-162-254: ~/Desktop/Exercise-Files/TASK-2 (HTTP)
File  Edit  View  Search  Terminal  Help
     S5 G 2:            0 (   0.000%)
      Total:          460
===============================================================================
Action Stats:
     Alerts:          164 ( 35.652%)
     Logged:          164 ( 35.652%)
     Passed:            0 (   0.000%)
Limits:
      Match:            0
      Queue:            0
        Log:            0
      Event:            0
      Alert:            0
Verdicts:
      Allow:          460 (100.000%)
      Block:            0 (   0.000%)
    Replace:            0 (   0.000%)
  Whitelist:            0 (   0.000%)
  Blacklist:            0 (   0.000%)
     Ignore:            0 (   0.000%)
      Retry:            0 (   0.000%)
===============================================================================
Snort exiting
ubuntu@ip-10-67-162-254:~/Desktop/Exercise-Files/TASK-2 (HTTP)$
```

2. Investigate the log file. What is the destination address of packet 63? **216.239.59.99**

```
snort -dvr snort.log.1738265409 -n 63
```

```
ubuntu@ip-10-67-162-254: ~/Desktop/Exercise-Files/TASK-2 (HTTP)
File  Edit  View  Search  Terminal  Help
WARNING: No preprocessors configured for policy 0.
05/13-10:17:10.295515 145.254.160.237:3371 -> 216.239.59.99:80
TCP TTL:128 TOS:0x0 ID:3917 IpLen:20 DgmLen:761 DF
***AP*** Seq: 0x36C21E28  Ack: 0x2E6B5384  Win: 0x2238  TcpLen: 20
47 45 54 20 2F 70 61 67 65 61 64 2F 61 64 73 3F  GET /pagead/ads?
63 6C 69 65 6E 74 3D 63 61 2D 70 75 62 2D 32 33  client=ca-pub-23
30 39 31 39 31 39 34 38 36 37 33 36 32 39 26 72  09191948673629&r
61 6E 64 6F 6D 3D 31 30 38 34 34 34 33 34 33 30  andom=1084443430
32 38 35 26 6C 6D 74 3D 31 30 38 32 34 36 37 30  285&lmt=10824670
32 30 26 66 6F 72 6D 61 74 3D 34 36 38 78 36 30  20&format=468x60
5F 61 73 26 6F 75 74 70 75 74 3D 68 74 6D 6C 26  _as&output=html&
75 72 6C 3D 68 74 74 70 25 33 41 25 32 46 25 32  url=http%3A%2F%2
46 77 77 77 2E 65 74 68 65 72 65 61 6C 2E 63 6F  Fwww.ethereal.co
6D 25 32 46 64 6F 77 6E 6C 6F 61 64 2E 68 74 6D  m%2Fdownload.htm
6C 26 63 6F 6C 6F 72 5F 62 67 3D 46 46 46 46 46  l&color_bg=FFFFF
46 26 63 6F 6C 6F 72 5F 74 65 78 74 3D 33 33 33  F&color_text=333
33 33 33 26 63 6F 6C 6F 72 5F 6C 69 6E 6B 3D 30  333&color_link=0
30 30 30 30 30 26 63 6F 6C 6F 72 5F 75 72 6C 3D  00000&color_url=
36 36 36 36 33 33 26 63 6F 6C 6F 72 5F 62 6F 72  666633&color_bor
64 65 72 3D 36 36 36 36 33 33 20 48 54 54 50 2F  der=666633 HTTP/
31 2E 31 0D 0A 48 6F 73 74 3A 20 70 61 67 65 61  1.1..Host: pagea
64 32 2E 67 6F 6F 67 6C 65 73 79 6E 64 69 63 61  d2.googlesyndica
74 69 6F 6E 2E 63 6F 6D 0D 0A 55 73 65 72 2D 41  tion.com..User-A
67 65 6E 74 3A 20 4D 6F 7A 69 6C 6C 61 2F 35 2E  gent: Mozilla/5.
```

3. Investigate the log file. What is the ACK number of packet 64? **0x2E6B5384**

```
snort -dvr snort.log.1738265409 -n 64
```

```
ubuntu@ip-10-66-184-195: ~/Desktop/Exercise-Files/TASK-2 (HTTP)
File  Edit  View  Search  Terminal  Help
WARNING: No preprocessors configured for policy 0.
05/13-10:17:10.295515 145.254.160.237:3371 -> 216.239.59.99:80
TCP TTL:128 TOS:0x0 ID:3917 IpLen:20 DgmLen:761 DF
***AP*** Seq: 0x36C21E28  Ack: 0x2E6B5384  Win: 0x2238  TcpLen: 20
47 45 54 20 2F 70 61 67 65 61 64 2F 61 64 73 3F  GET /pagead/ads?
63 6C 69 65 6E 74 3D 63 61 2D 70 75 62 2D 32 33  client=ca-pub-23
30 39 31 39 31 39 34 38 36 37 33 36 32 39 26 72  091919486736 29&r
61 6E 64 6F 6D 3D 31 30 38 34 34 34 33 34 33 30  andom=1084443430
32 38 35 26 6C 6D 74 3D 31 30 38 32 34 36 37 30  285&lmt=10824670
32 30 26 66 6F 72 6D 61 74 3D 34 36 38 78 36 30  20&format=468x60
5F 61 73 26 6F 75 74 70 75 74 3D 68 74 6D 6C 26  _as&output=html&
75 72 6C 3D 68 74 74 70 25 33 41 25 32 46 25 32  url=http%3A%2F%2
46 77 77 77 2E 65 74 68 65 72 65 61 6C 2E 63 6F  Fwww.ethereal.co
6D 25 32 46 64 6F 77 6E 6C 6F 61 64 2E 68 74 6D  m%2Fdownload.htm
6C 26 63 6F 6C 6F 72 5F 62 67 3D 46 46 46 46 46  l&color_bg=FFFFF
46 26 63 6F 6C 6F 72 5F 74 65 78 74 3D 33 33 33  F&color_text=333
33 33 33 26 63 6F 6C 6F 72 5F 6C 69 6E 6B 3D 30  333&color_link=0
30 30 30 30 30 26 63 6F 6C 6F 72 5F 75 72 6C 3D  00000&color_url=
36 36 36 36 33 33 26 63 6F 6C 6F 72 5F 62 6F 72  666633&color_bor
64 65 72 3D 36 36 36 36 33 33 20 48 54 54 50 2F  der=666633 HTTP/
31 2E 31 0D 0A 48 6F 73 74 3A 20 70 61 67 65 61  1.1..Host: pagea
64 32 2E 67 6F 6F 67 6C 65 73 79 6E 64 69 63 61  d2.googlesyndica
74 69 6F 6E 2E 63 6F 6D 0D 0A 55 73 65 72 2D 41  tion.com..User-A
67 65 6E 74 3A 20 4D 6F 7A 69 6C 6C 61 2F 35 2E  gent: Mozilla/5.
```

4. Investigate the log file. What is the SEQ number of packet 62? **0x36C21E28**

```
snort -dvr snort.log.1738265409 -n 62
```

```
ubuntu@ip-10-66-184-195: ~/Desktop/Exercise-Files/TASK-2 (HTTP)
File  Edit  View  Search  Terminal  Help
05/13-10:17:10.295515 145.254.160.237:3371 -> 216.239.59.99:80
TCP TTL:128 TOS:0x0 ID:3917 IpLen:20 DgmLen:761 DF
***AP*** Seq: 0x36C21E28  Ack: 0x2E6B5384  Win: 0x2238  TcpLen: 20
47 45 54 20 2F 70 61 67 65 61 64 2F 61 64 73 3F  GET /pagead/ads?
63 6C 69 65 6E 74 3D 63 61 2D 70 75 62 2D 32 33  client=ca-pub-23
30 39 31 39 31 39 34 38 36 37 33 36 32 39 26 72  091919486736 29&r
61 6E 64 6F 6D 3D 31 30 38 34 34 34 33 34 33 30  andom=1084443430
32 38 35 26 6C 6D 74 3D 31 30 38 32 34 36 37 30  285&lmt=10824670
32 30 26 66 6F 72 6D 61 74 3D 34 36 38 78 36 30  20&format=468x60
5F 61 73 26 6F 75 74 70 75 74 3D 68 74 6D 6C 26  _as&output=html&
75 72 6C 3D 68 74 74 70 25 33 41 25 32 46 25 32  url=http%3A%2F%2
46 77 77 77 2E 65 74 68 65 72 65 61 6C 2E 63 6F  Fwww.ethereal.co
6D 25 32 46 64 6F 77 6E 6C 6F 61 64 2E 68 74 6D  m%2Fdownload.htm
6C 26 63 6F 6C 6F 72 5F 62 67 3D 46 46 46 46 46  l&color_bg=FFFFF
46 26 63 6F 6C 6F 72 5F 74 65 78 74 3D 33 33 33  F&color_text=333
33 33 33 26 63 6F 6C 6F 72 5F 6C 69 6E 6B 3D 30  333&color_link=0
30 30 30 30 30 26 63 6F 6C 6F 72 5F 75 72 6C 3D  00000&color_url=
36 36 36 36 33 33 26 63 6F 6C 6F 72 5F 62 6F 72  666633&color_bor
64 65 72 3D 36 36 36 36 33 33 20 48 54 54 50 2F  der=666633 HTTP/
31 2E 31 0D 0A 48 6F 73 74 3A 20 70 61 67 65 61  1.1..Host: pagea
64 32 2E 67 6F 6F 67 6C 65 73 79 6E 64 69 63 61  d2.googlesyndica
74 69 6F 6E 2E 63 6F 6D 0D 0A 55 73 65 72 2D 41  tion.com..User-A
67 65 6E 74 3A 20 4D 6F 7A 69 6C 6C 61 2F 35 2E  gent: Mozilla/5.
30 20 28 57 69 6E 64 6F 77 73 3B 20 55 3B 20 57  0 (Windows; U; W
```

5. Investigate the log file. What is the TTL of packet 65? **128**

```
snort -dvr snort.log.1738265409 -n 65
```

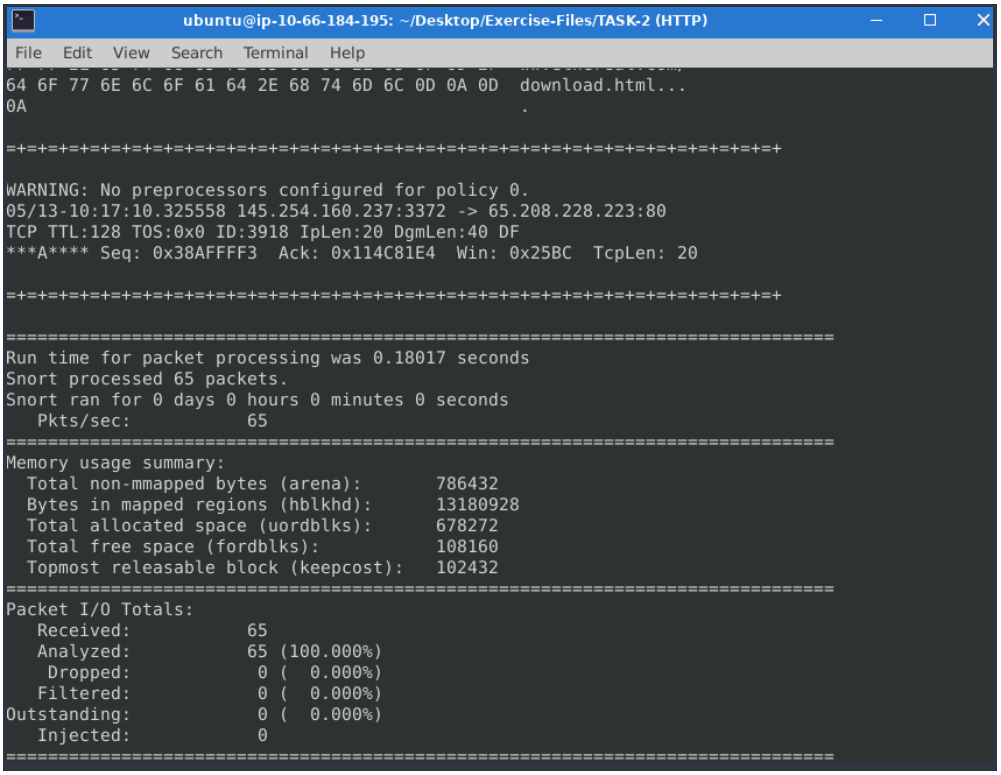
6. Investigate the log file. What is the source IP of packet 65? **145.254.160.237**

```
snort -dvr snort.log.1738265409 -n 65
```

7. Investigate the log file. What is the source port of packet 65? **3372**

```
snort -dvr snort.log.1738265409 -n 65
```

ALL OF THESE USE THE SAME THING:



```
ubuntu@ip-10-66-184-195: ~/Desktop/Exercise-Files/TASK-2 (HTTP)
File  Edit  View  Search  Terminal  Help
64 6F 77 6E 6C 6F 61 64 2E 68 74 6D 6C 0D 0A 0D  download.html...
0A                                               .

=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
05/13-10:17:10.325558 145.254.160.237:3372 -> 65.208.228.223:80
TCP TTL:128 TOS:0x0 ID:3918 IpLen:20 DgmLen:40 DF
***A**** Seq: 0x38AFFFF3  Ack: 0x114C81E4  Win: 0x25BC  TcpLen: 20

=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

===============================================================================
Run time for packet processing was 0.18017 seconds
Snort processed 65 packets.
Snort ran for 0 days 0 hours 0 minutes 0 seconds
   Pkts/sec:           65
===============================================================================
Memory usage summary:
  Total non-mmapped bytes (arena):       786432
  Bytes in mapped regions (hblkhd):      13180928
  Total allocated space (uordblks):      678272
  Total free space (fordblks):           108160
  Topmost releasable block (keepcost):   102432
===============================================================================
Packet I/O Totals:
   Received:           65
   Analyzed:           65 (100.000%)
    Dropped:            0 (  0.000%)
   Filtered:            0 (  0.000%)
Outstanding:            0 (  0.000%)
   Injected:            0
===============================================================================
```

IDS Rules for FTP traffic

1. Write a single rule to detect "all TCP port 21" traffic in the given pcap. What is the number of detected packets? **307**

   Step 1: Open the ‘Local Rules’ file
   Step 2: Inside ‘Local Rules’, `alert tcp any 21 -> any any (msg:"FTP traffic from source"; sid:100001; rev:1;)`
   `alert tcp any any -> any 21 (msg:"FTP traffic to destination"; sid:100002; rev:1;)` (this rule is a 2-way handshake between the source IP and the destination IP. There’s something related to Port 21…)
   Step 3:Inside Terminal, `snort -c local.rules -A full -l . -r ftp-png-gif.pcap`

```
ubuntu@ip-10-66-145-94: ~/Desktop/Exercise-Files/TASK-3 (FTP)
File  Edit  View  Search  Terminal  Help
     S5 G 2:            0 (  0.000%)
      Total:          421
==================================================================================
Action Stats:
     Alerts:          307 ( 72.922%)
     Logged:          307 ( 72.922%)
     Passed:            0 (  0.000%)
Limits:
      Match:            0
      Queue:            0
        Log:            0
      Event:            0
      Alert:            0
Verdicts:
      Allow:          421 (100.000%)
      Block:            0 (  0.000%)
    Replace:            0 (  0.000%)
  Whitelist:            0 (  0.000%)
  Blacklist:            0 (  0.000%)
     Ignore:            0 (  0.000%)
      Retry:            0 (  0.000%)
==================================================================================
Snort exiting
ubuntu@ip-10-66-145-94:~/Desktop/Exercise-Files/TASK-3 (FTP)$
```

2. Investigate the log file. What is the FTP service name? **Microsoft FTP Service**

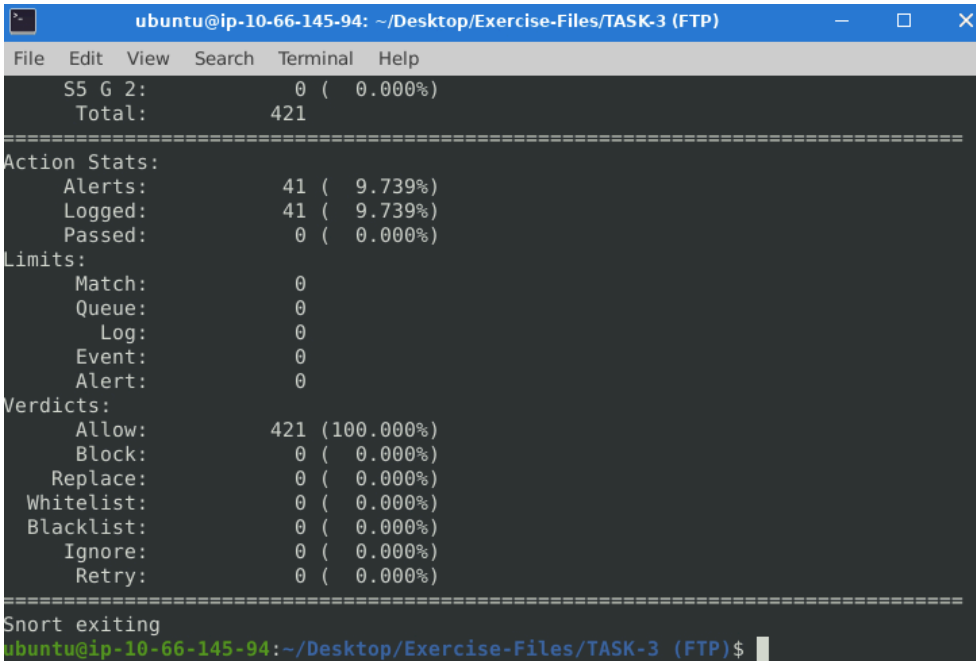
```
snort -dvr snort.log.1772294791
```

```
ubuntu@ip-10-66-145-94: ~/Desktop/Exercise-Files/TASK-3 (FTP)
File  Edit  View  Search  Terminal  Help
WARNING: No preprocessors configured for policy 0.
01/04-10:19:34.008856 192.168.75.132:21 -> 192.168.75.1:18157
TCP TTL:128 TOS:0x0 ID:1696 IpLen:20 DgmLen:79 DF
***AP*** Seq: 0x93FDAA43  Ack: 0xE9CEC219  Win: 0xFAF0  TcpLen: 32
TCP Options (3) => NOP NOP TS: 13955 7457661
32 32 30 20 4D 69 63 72 6F 73 6F 66 74 20 46 54  220 Microsoft FT
50 20 53 65 72 76 69 63 65 0D 0A                 P Service..

=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
01/04-10:19:34.018155 192.168.75.1:18158 -> 192.168.75.132:21
TCP TTL:128 TOS:0x0 ID:2434 IpLen:20 DgmLen:60 DF
******S* Seq: 0x9699364F  Ack: 0x0  Win: 0x2000  TcpLen: 40
TCP Options (5) => MSS: 1460 NOP WS: 2 SackOK TS: 7457662 0

=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
01/04-10:19:34.018411 192.168.75.132:21 -> 192.168.75.1:18158
TCP TTL:128 TOS:0x0 ID:1697 IpLen:20 DgmLen:64
***A**S* Seq: 0x16A54E38  Ack: 0x96993650  Win: 0xFAF0  TcpLen: 44
TCP Options (9) => MSS: 1460 NOP WS: 0 NOP NOP TS: 0 0 NOP NOP SackOK
```

3. Write a rule to detect failed FTP login attempts in the given pcap. What is the number of detected packets? **41**

Inside ‘Local Rules’, `alert tcp any 21 -> any any (content:"530";msg:"failed FTP login";sid:100001; rev:1;)` (this rule catches failed authentication attempts on an FTP server)

Inside Terminal, `snort -c local.rules -A full -l . -r ftp-png-gif.pcap`



4. Write a rule to detect successful FTP logins in the given pcap. What is the number of detected packets? **1**

Inside ‘Local Rules’, `alert tcp any 21 -> any any (content:"230";msg:"successful FTP login";sid:100001; rev:1;)` (this rule catches successful authentication attempts on an FTP server)

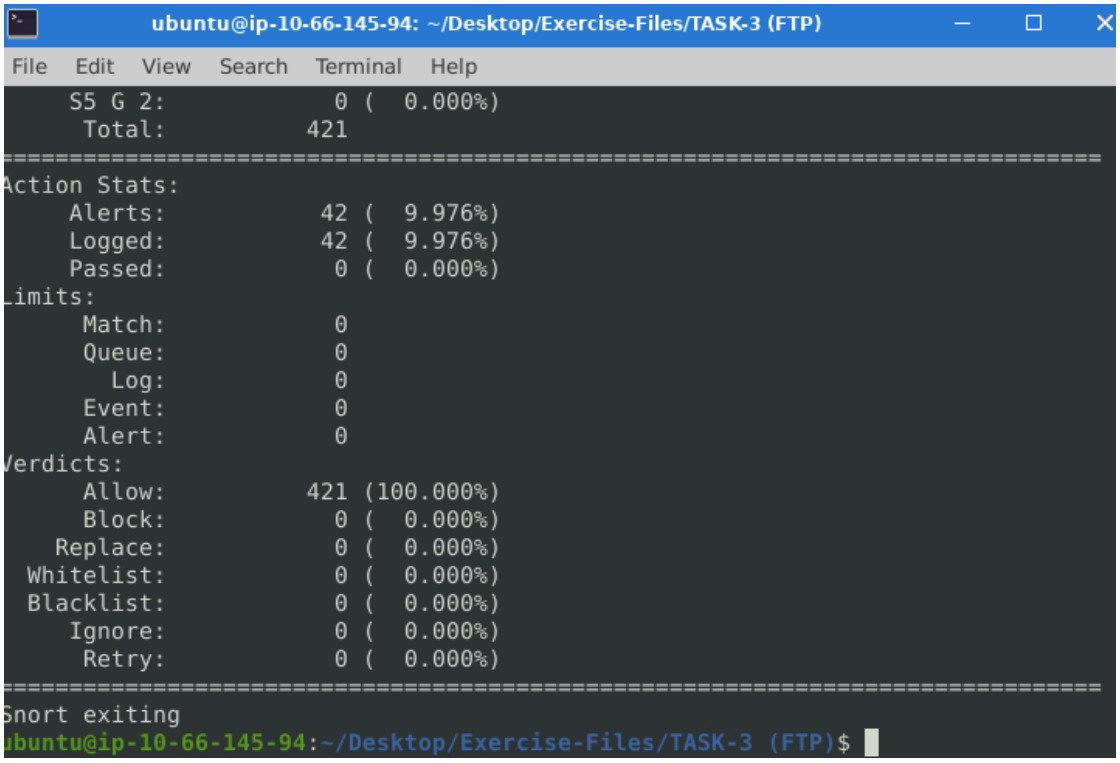
Inside Terminal, `snort -c local.rules -A full -l . -r ftp-png-gif.pcap`

```
ubuntu@ip-10-66-145-94: ~/Desktop/Exercise-Files/TASK-3 (FTP)
File  Edit  View  Search  Terminal  Help
     S5 G 2:             0 (   0.000%)
      Total:           421
===============================================================================
Action Stats:
     Alerts:             1 (   0.238%)
     Logged:             1 (   0.238%)
     Passed:             0 (   0.000%)
Limits:
      Match:             0
      Queue:             0
        Log:             0
      Event:             0
      Alert:             0
Verdicts:
      Allow:           421 (100.000%)
      Block:             0 (   0.000%)
    Replace:             0 (   0.000%)
  Whitelist:             0 (   0.000%)
  Blacklist:             0 (   0.000%)
     Ignore:             0 (   0.000%)
      Retry:             0 (   0.000%)
===============================================================================
Snort exiting
ubuntu@ip-10-66-145-94:~/Desktop/Exercise-Files/TASK-3 (FTP)$
```

4. Write a rule to detect FTP login attempts with a valid username but no password entered yet. What is the number of detected packets?

Inside 'Local Rules', `alert tcp any 21 -> any any (content:"331";msg:"username OK, need password";sid:100001; rev:1;)` (this rule detects when an FTP server acknowledges a username and asks the user for a password.)

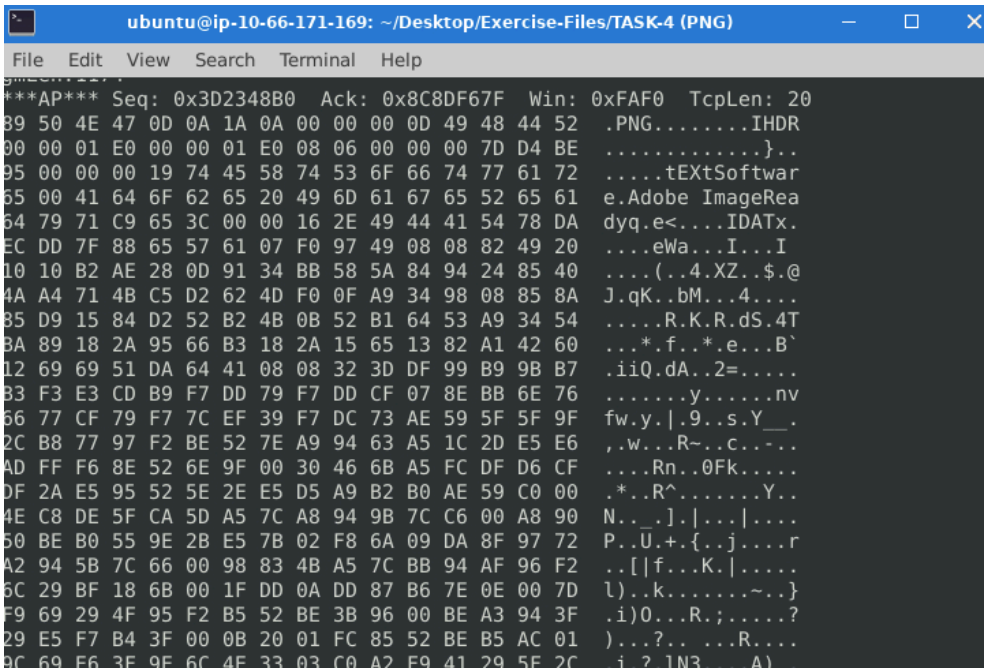
Inside Terminal, `snort -c local.rules -A full -l . -r ftp-png-gif.pcap`



IDS Rules for PNG files

1. Write a rule to detect the PNG file in the given pcap. Investigate the logs and identify the software name embedded in the packet.

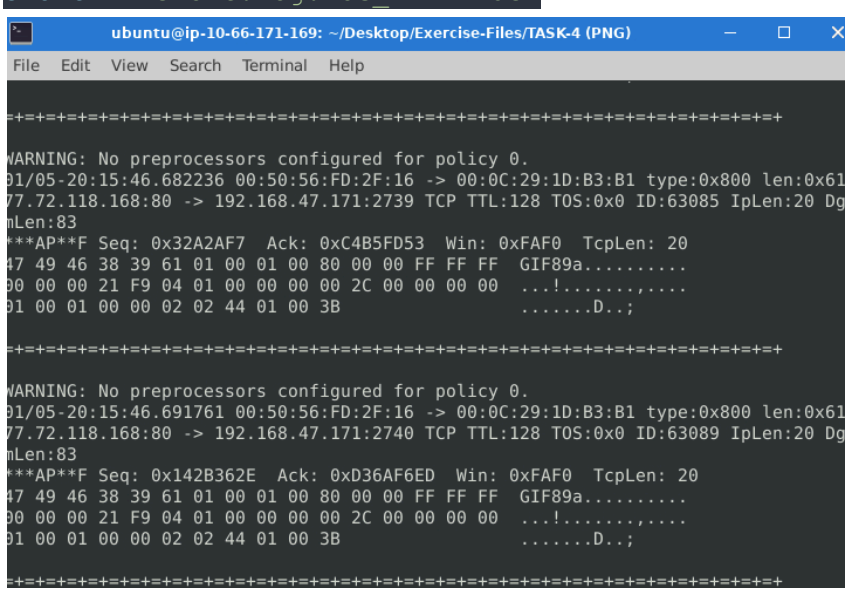
Inside 'Local Rules', alert tcp any any <> any any (msg:"PNG file detected"; content:"|89 50 4E 47 0D 0A 1A 0A|"; sid:1000001; rev:1;) (this rule detects when a PNG is found.)
Inside Terminal, `snort -c local.rules -A full -l . -r ftp-png-gif.pcap` then `snort -r snort.log.LOG_NAME -dev`



2. Write a rule to detect the GIF file in the given pcap. Investigate the logs and identify the image format embedded in the packet.

Inside 'Local Rules', alert tcp any any <> any any (msg:"GIF file detected"; content:"|47 49 46 38 39 61|"; sid:1000001; rev:1;) (this rule detects when a GIF is found.)

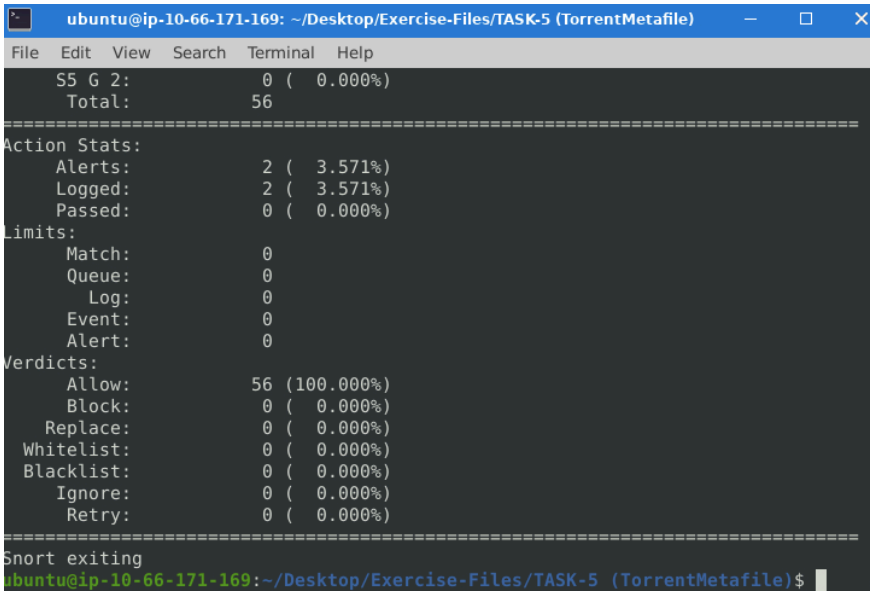
Inside Terminal, `snort -c local.rules -A full -l . -r ftp-png-gif.pcap` then `snort -r snort.log.LOG_NAME -dev`



IDS Rules for torrent metafiles traffic

1. Write a rule to detect the torrent metafile in the given pcap. What is the number of detected packets? **2**

Inside 'Local Rules', alert tcp any any <> any any (msg:"Torrent file detected"; content:"torrent"; sid:1000001; rev:1;) (this rule detects the presence of BitTorrent-related traffic by looking for the plain-text string `"torrent"` inside any TCP packet.)
Inside Terminal, `snort -c local.rules -A full -l . -r ftp-png-gif.pcap` then



2. Investigate the log/alarm files. What is the name of the torrent application? **bittorrent**

```
snort -r snort.log.LOG_NAME -dev
```

```
ubuntu@ip-10-66-171-169: ~/Desktop/Exercise-Files/TASK-5 (TorrentMetafile)
File  Edit  View  Search  Terminal  Help
***AP*** Seq: 0xE9A59016  Ack: 0xED0BC172  Win: 0x2238  TcpLen: 20
47 45 54 20 2F 61 6E 6E 6F 75 6E 63 65 3F 69 6E  GET /announce?in
66 6F 5F 68 61 73 68 3D 25 30 31 64 25 46 45 25  fo_hash=%01d%FE%
37 45 25 46 31 25 31 30 25 35 43 57 76 41 70 25  7E%F1%10%5CWvAp%
45 44 25 46 36 25 30 33 25 43 34 39 25 44 36 42  ED%F6%03%C49%D6B
25 31 34 25 46 31 26 70 65 65 72 5F 69 64 3D 25  %14%F1&peer_id=%
42 38 6A 73 25 37 46 25 45 38 25 30 43 25 41 46  B8js%7F%E8%0C%AF
68 25 30 32 59 25 39 36 37 25 32 34 65 25 32 37  h%02Y%967%24e%27
56 25 45 45 4D 25 31 36 25 35 42 26 70 6F 72 74  V%EEM%16%5B&port
3D 34 31 37 33 30 26 75 70 6C 6F 61 64 65 64 3D  =41730&uploaded=
30 26 64 6F 77 6E 6C 6F 61 64 65 64 3D 30 26 6C  0&downloaded=0&l
65 66 74 3D 33 37 36 37 38 36 39 26 63 6F 6D 70  eft=3767869&comp
61 63 74 3D 31 26 69 70 3D 31 32 37 2E 30 2E 30  act=1&ip=127.0.0
2E 31 26 65 76 65 6E 74 3D 73 74 61 72 74 65 64  .1&event=started
20 48 54 54 50 2F 31 2E 31 0D 0A 41 63 63 65 70   HTTP/1.1..Accep
74 3A 20 61 70 70 6C 69 63 61 74 69 6F 6E 2F 78  t: application/x
2D 62 69 74 74 6F 72 72 65 6E 74 0D 0A 41 63 63  -bittorrent..Acc
65 70 74 2D 45 6E 63 6F 64 69 6E 67 3A 20 67 7A  ept-Encoding: gz
69 70 0D 0A 55 73 65 72 2D 41 67 65 6E 74 3A 20  ip..User-Agent:
52 41 5A 41 20 32 2E 31 2E 30 2E 30 0D 0A 48 6F  RAZA 2.1.0.0..Ho
73 74 3A 20 74 72 61 63 6B 65 72 32 2E 74 6F 72  st: tracker2.tor
72 65 6E 74 62 6F 78 2E 63 6F 6D 3A 32 37 31 30  rentbox.com:2710
0D 0A 43 6F 6E 6E 65 63 74 69 6F 6E 3A 20 4B 65  ..Connection: Ke
65 70 2D 41 6C 69 76 65 0D 0A 0D 0A              ep-Alive....
```

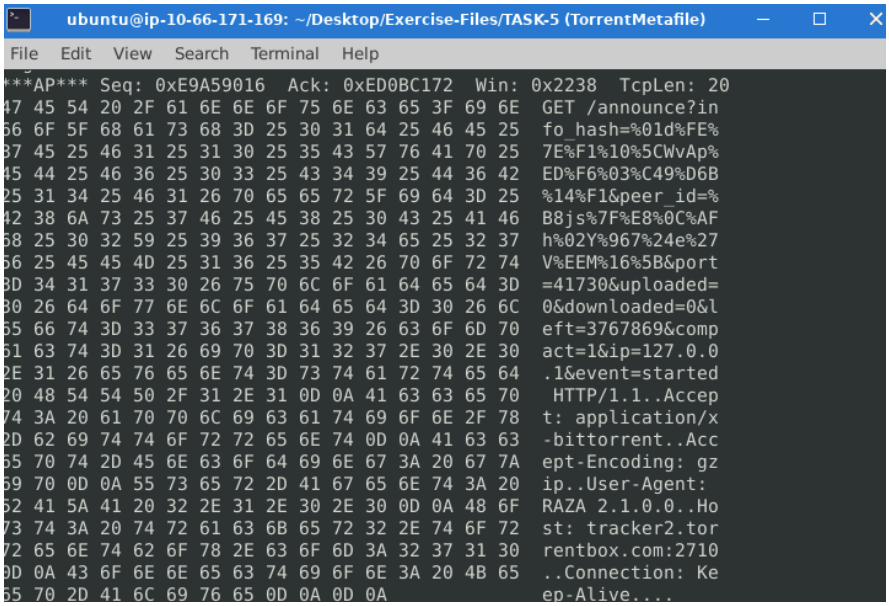
3. Investigate the log/alarm files. What is the MIME (Multipurpose Internet Mail Extensions) type of the torrent metafile? **application/x-bittorrent**

```
snort -r snort.log.LOG_NAME -dev
```


```
ubuntu@ip-10-66-171-169: ~/Desktop/Exercise-Files/TASK-5 (TorrentMetafile)
File  Edit  View  Search  Terminal  Help
***AP*** Seq: 0xE9A59016  Ack: 0xED0BC172  Win: 0x2238  TcpLen: 20
47 45 54 20 2F 61 6E 6E 6F 75 6E 63 65 3F 69 6E  GET /announce?in
66 6F 5F 68 61 73 68 3D 25 30 31 64 25 46 45 25  fo_hash=%01d%FE%
37 45 25 46 31 25 31 30 25 35 43 57 76 41 70 25  7E%F1%10%5CWvAp%
45 44 25 46 36 25 30 33 25 43 34 39 25 44 36 42  ED%F6%03%C49%D6B
25 31 34 25 46 31 26 70 65 65 72 5F 69 64 3D 25  %14%F1&peer_id=%
42 38 6A 73 25 37 46 25 45 38 25 30 43 25 41 46  B8js%7F%E8%0C%AF
68 25 30 32 59 25 39 36 37 25 32 34 65 25 32 37  h%02Y%967%24e%27
56 25 45 45 4D 25 31 36 25 35 42 26 70 6F 72 74  V%EEM%16%5B&port
3D 34 31 37 33 30 26 75 70 6C 6F 61 64 65 64 3D  =41730&uploaded=
30 26 64 6F 77 6E 6C 6F 61 64 65 64 3D 30 26 6C  0&downloaded=0&l
65 66 74 3D 33 37 36 37 38 36 39 26 63 6F 6D 70  eft=3767869&comp
61 63 74 3D 31 26 69 70 3D 31 32 37 2E 30 2E 30  act=1&ip=127.0.0
2E 31 26 65 76 65 6E 74 3D 73 74 61 72 74 65 64  .1&event=started
20 48 54 54 50 2F 31 2E 31 0D 0A 41 63 63 65 70   HTTP/1.1..Accep
74 3A 20 61 70 70 6C 69 63 61 74 69 6F 6E 2F 78  t: application/x
2D 62 69 74 74 6F 72 72 65 6E 74 0D 0A 41 63 63  -bittorrent..Acc
65 70 74 2D 45 6E 63 6F 64 69 6E 67 3A 20 67 7A  ept-Encoding: gz
69 70 0D 0A 55 73 65 72 2D 41 67 65 6E 74 3A 20  ip..User-Agent:
52 41 5A 41 20 32 2E 31 2E 30 2E 30 0D 0A 48 6F  RAZA 2.1.0.0..Ho
73 74 3A 20 74 72 61 63 6B 65 72 32 2E 74 6F 72  st: tracker2.tor
72 65 6E 74 62 6F 78 2E 63 6F 6D 3A 32 37 31 30  rentbox.com:2710
0D 0A 43 6F 6E 6E 65 63 74 69 6F 6E 3A 20 4B 65  ..Connection: Ke
65 70 2D 41 6C 69 76 65 0D 0A 0D 0A              ep-Alive....
```


4. Investigate the log/alarm files. What is the hostname of the torrent metafile? **tracker2.torrentbox.com**

```
snort -r snort.log.LOG_NAME -dev
```



```
ubuntu@ip-10-66-171-169: ~/Desktop/Exercise-Files/TASK-5 (TorrentMetafile)
File  Edit  View  Search  Terminal  Help
***AP*** Seq: 0xE9A59016  Ack: 0xED0BC172  Win: 0x2238  TcpLen: 20
47 45 54 20 2F 61 6E 6E 6F 75 6E 63 65 3F 69 6E  GET /announce?in
66 6F 5F 68 61 73 68 3D 25 30 31 64 25 46 45 25  fo_hash=%01d%FE%
37 45 25 46 31 25 31 30 25 35 43 57 76 41 70 25  7E%F1%10%5CWvAp%
45 44 25 46 36 25 30 33 25 43 34 39 25 44 36 42  ED%F6%03%C49%D6B
25 31 34 25 46 31 26 70 65 65 72 5F 69 64 3D 25  %14%F1&peer_id=%
42 38 6A 73 25 37 46 25 45 38 25 30 43 25 41 46  B8js%7F%E8%0C%AF
68 25 30 32 59 25 39 36 37 25 32 34 65 25 32 37  h%02Y%967%24e%27
56 25 45 45 4D 25 31 36 25 35 42 26 70 6F 72 74  V%EEM%16%5B&port
3D 34 31 37 33 30 26 75 70 6C 6F 61 64 65 64 3D  =41730&uploaded=
30 26 64 6F 77 6E 6C 6F 61 64 65 64 3D 30 26 6C  0&downloaded=0&l
65 66 74 3D 33 37 36 37 38 36 39 26 63 6F 6D 70  eft=3767869&comp
61 63 74 3D 31 26 69 70 3D 31 32 37 2E 30 2E 30  act=1&ip=127.0.0
2E 31 26 65 76 65 6E 74 3D 73 74 61 72 74 65 64  .1&event=started
20 48 54 54 50 2F 31 2E 31 0D 0A 41 63 63 65 70   HTTP/1.1..Accep
74 3A 20 61 70 70 6C 69 63 61 74 69 6F 6E 2F 78  t: application/x
2D 62 69 74 74 6F 72 72 65 6E 74 0D 0A 41 63 63  -bittorrent..Acc
65 70 74 2D 45 6E 63 6F 64 69 6E 67 3A 20 67 7A  ept-Encoding: gz
69 70 0D 0A 55 73 65 72 2D 41 67 65 6E 74 3A 20  ip..User-Agent:
52 41 5A 41 20 32 2E 31 2E 30 2E 30 0D 0A 48 6F  RAZA 2.1.0.0..Ho
73 74 3A 20 74 72 61 63 6B 65 72 32 2E 74 6F 72  st: tracker2.tor
72 65 6E 74 62 6F 78 2E 63 6F 6D 3A 32 37 31 30  rentbox.com:2710
0D 0A 43 6F 6E 6E 65 63 74 69 6F 6E 3A 20 4B 65  ..Connection: Ke
65 70 2D 41 6C 69 76 65 0D 0A 0D 0A              ep-Alive....
```

Reflection

In the lab, any and all Snort rules triggered during the test. That was necessary in order for me to complete the questions. I don't know what would have happened if the rules hadn't been activated (I'll use this word going forward). The only problem that I know of when it comes to Snort rules is that they are written in complete gibberish - I couldn't understand them. Therefore, I had to ask Google Gemini to tell me what those rules do.

In hindsight, I should've learned a lot more about the Snort rules before completing the lab. Oh well. At least now I know how to create Snort rules.